
Neurosymbolic Supervision with Explicit Evidence: A Cold Repair Study

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Abstract

1 Coding supervisors must separate source-linked obligations from operational de-
2 cisions and from the evidence that may admit a candidate. We report a frozen
3 eight-family comparison of that bounded composition: whether adding native
4 source-linked semantic context to matched public raw context changes useful cold
5 repair under a fixed resource policy. The retained design is 32 fixed A/B local-
6 cold cells across eight convenience-selected repository families and two nested
7 repetitions (104729, 130363), an order-preserving subset of an original 192-cell
8 plan. Useful completion is 9/32 (arm A 5/16, arm B 4/16). The mean family B-
9 minus-A useful fraction is -0.0625 , with a descriptive conditional 95% percentile
10 interval $[-0.1875, 0]$ from 20,000 whole-family bootstrap resamples. Sixteen scor-
11 ing invocations produced fifteen completed cold validations; one additional scored
12 cell had incomplete hidden collection. Fourteen cells hit child deadlines and two
13 were known proposal failures. All 32 cells remain in the denominator. C/D routing,
14 warm caches, reuse, publication-comparison efficacy, and original sixteen-family
15 inferential objectives were withdrawn before outcomes. Human semantic fidelity
16 was not measured. Settled provider charges remain unavailable. The result is
17 descriptive of these eight families; it does not establish superiority, noninferiority,
18 or population-level coverage.

19 1 Introduction

20 A coding supervisor must distinguish what an edit changes, what evidence applies to the resulting
21 program, and who may accept the transition. A stored passing result can concern an earlier source
22 revision; a completed task can record an operational decision without establishing a behavioral
23 property. Repository-agent interfaces and test-validation benchmarks make the boundary between
24 proposals and checks central to useful repair [6, 3]. Our starting point is to make each observation’s
25 subject, scope, and authority explicit.

26 The architecture separates semantic state, which records source-linked facts and unresolved obliga-
27 tions, from operational state, which records work and decisions, and artifact storage, which retains
28 candidate bytes and evidence. A proposed transition identifies evidence for its declared acceptance
29 conditions. Unknown dependencies and unsupported translations remain visible obligations. Neither
30 persistence nor a cryptographic commitment turns an observation into proof of an unstated property.

31 The contribution is a design for these evidence boundaries, conditional acceptance and reuse argu-
32 ments, and a bounded implementation comparison. The empirical question is whether adding native
33 source-linked semantic context to matched public raw context changes useful cold repair under a
34 fixed resource policy. The retained design contains eight repository families, two arms, and two
35 nested repetitions per arm. It tests the combined context mechanism, rather than every component of

36 the intended supervision loop. Symbolic routing, warm-cache reuse, publication efficacy, and native
37 proving performance require separate evidence.

38 A returned proposal is not a successful repair. Useful completion requires actual nonempty public and
39 hidden checks to pass within the measured resource rules; provider failures, timeouts, and incomplete
40 validation stay in the fixed denominator. This separation also governs the reporting of setup work,
41 earlier pilot results, runtime interruptions, and unknown costs.

42 2 Method and trusted assumptions

43 **Separate evidence classes.** Semantic state links facts, dependencies, contracts, and unresolved
44 obligations to source. Operational state records goals, tasks, attempts, leases, and decisions. Artifact
45 storage preserves candidate bytes and accepted manifests. An evidence item binds a subject to a
46 declared statement, checker, environment, and provenance. A passing test concerns the finite checks
47 actually executed; a formal derivation concerns its encoded relation and premises; a grant concerns
48 an actor’s permission to act now. These classes cannot substitute for one another. In particular, a
49 signature binds a record to a key under cryptographic assumptions. It establishes neither checker
50 adequacy nor the honesty of an observation.

51 **Applicable evidence.** The intended loop derives obligations from current source, chooses an
52 admissible analysis or proposal route, checks an isolated candidate, and admits only evidence
53 applicable to that candidate. Translation validation motivates checking a particular conversion against
54 its stated relation [5]; a source-linked record alone is not such validation. Abstract facts likewise
55 depend on an approximation relation [1]. A typed or content-addressed representation does not imply
56 full source-language soundness.

57 **Conditional acceptance and reuse.** If publication succeeds only after checking the current parent,
58 authorized scope, complete requirement manifest, admitted evidence, and expected-parent compare-
59 and-swap, each successful successor satisfies those gates. Given a valid initial root and exclusive
60 use of that publication operation, induction preserves the declared manifest policy. This does not
61 establish enforcement by unexamined write paths or atomicity of external effects.

62 For test reuse, suppose the observation of test t under profile θ factors through a complete dependency
63 projection:

$$\text{Obs}_\theta(t, s) = f_\theta(\pi_D(s)).$$

64 If f_θ is deterministic on every admitted state, equal projections imply equal observations. Com-
65 pleteness and determinism are substantive premises; hashing an incomplete dependency list, opaque
66 fixture, or uncontrolled effect does not discharge them. Appendix A preserves the full conditional,
67 unmachine-checked statements and the distinction between behavioral acceptance, action permission,
68 and publication.

69 **Executed scope.** The final comparison exercises native context preparation, model proposals,
70 candidate custody and AI inspection, and separate cold scoring. Earlier constructed controls and a
71 local guard-before-write witness test named implementation boundaries; they do not establish that
72 the complete intended loop ran as a live benchmark. The retained proof profile lacks the admitted
73 backend, key provenance, and exact statement binding required for native proof admission. Neither
74 native proving performance nor empirical reuse benefit is inferred from these arguments.

75 3 Frozen comparison

76 **Population and treatment.** The eight recruited families are Tomlkit, installer, Tornado, more-
77 itertools, charset-normalizer, iniconfig, wheel, and Jinja. They are a convenience sample, not a
78 representative sample of Python repositories. Two arms and two nested repetitions yield 32 fixed
79 cells, 16 per arm. Repetition identifiers 104729 and 130363 identify design positions; they were not
80 served to the model API as random seeds.

81 Before final outcomes, the A/B local-cold comparison was retained as an order-preserving subset
82 of the original 192-cell plan. The remaining 160 factor cells were withdrawn, and eight additional
83 target families were not recruited. C/D routing, warm caches, test reuse, publication efficacy, the

84 original sixteen-family inferential objectives, and D/A promotion comparisons remain outside this
85 experiment. The narrowed comparison is descriptive.

86 Arm A receives the selected public raw context. Arm B preserves the matched raw selection and
87 adds actual native source-linked semantic context. Sixteen exact family/arm request commitments
88 precede final generation; each arm’s two repetitions use its fixed request. This treatment changes the
89 combined request and cannot isolate semantic subcomponents or prompt compression. Public source
90 and native preparation remain separate from confidential acceptance tests.

91 **One proposal and independent cold execution.** Each cell permits one Grok 4.6 proposal POST,
92 requesting at most 4,096 output tokens through a strict structured edit response. There is no scientific
93 fallback, retry, served seed, or reasoning-effort parameter. A failed request consumes its original cell
94 rather than creating a replacement. The frozen order is executed sequentially.

95 Proposal execution uses one CPU, 2 GiB, and a 32-process limit. The nominal gateway and HTTP-
96 wrapper bounds are 600 seconds; the child bound is 595 seconds and the socket timeout is 590 seconds.
97 Compliance depends on actual measured intervals. A configured limit, missing measurement, or
98 nonfinite value is not evidence that the budget was met.

99 The operator retains exact candidate bytes and a specific AI review before eligible scoring. That
100 procedural review supplies no independent human annotation. Scoring runs separately in a cold,
101 network-disabled container with one CPU, 2 GiB, and a 128-process limit, bounded by 120 host
102 seconds and 115 child seconds. The scorer’s process limit was fixed prospectively for both arms and
103 all eight families because unchanged public concurrency tests require up to 100 threads. The earlier
104 32-process public-baseline failure remains recorded; proposal and native-preparation limits remain
105 32.

106 Useful completion requires an admitted candidate, nonempty public and hidden checks all passing,
107 and measured resource compliance. Failed, refused, expired, unscored, or incomplete cells receive no
108 useful-completion credit. Unknown states remain explicit and prevent a complete-cohort analysis until
109 their terminal accounting is resolved. An import or collection error with no hidden assertions executed
110 is incomplete validation, not an observed hidden behavioral assertion failure. Separate scoring and
111 sealed records do not establish comprehensive resistance to adversarial scorer manipulation.

112 **Interruption and provenance.** The original resource lease ended during cell 7. Its request, failure,
113 cleanup, and unknown termination-sender evidence remain retained. A separately admitted managed
114 lease and continuation runtime govern cells 8–32 without replaying the consumed prefix or changing
115 requests, model, order, scorer, limits, or analysis. Both runtime source versions remain explicit. The
116 lease is a cooperative allocation, not an exclusive physical resource reservation; per-phase process
117 boundaries apply the configured resource limits.

118 Continuation checks add work around effect admission and receipt signing. The signed gateway
119 interval ends before signing; whole operator invocation time extends through subprocess completion
120 and output flush/synchronization. These distinct intervals must retain their endpoints rather than
121 being presented as a single total host clock.

122 4 Analysis and evidence separation

123 For family f , arm a , and repetition r , let Y_{far} indicate admitted useful completion. Average the two
124 fixed repetitions within each family and arm: $\bar{Y}_{fa} = (Y_{fa1} + Y_{fa2})/2$. The retained contrast is

$$\hat{\Delta} = \frac{1}{8} \sum_{f=1}^8 (\bar{Y}_{fB} - \bar{Y}_{fA}).$$

125 Families are the independent analysis units; 32 nested cells are not 32 independent repositories.
126 A descriptive robust-family outcome additionally requires both repetitions to succeed and is not a
127 promotion decision.

128 The frozen bootstrap draws eight whole family blocks with replacement, sharing each draw across
129 arms and repetitions. It uses 20,000 resamples, Python 3.12 `random.Random` with seed 20260911,
130 lexicographically ordered family identifiers, and exactly eight `randrange(8)` draws per resample.

131 The 95% percentile endpoints use linear interpolation at $(B - 1)p$ for $p \in \{0.025, 0.975\}$ and
132 $B = 20,000$. These intervals are conditional, descriptive summaries of eight convenience-selected
133 families, not population-level coverage or superiority/noninferiority decisions.

134 Failures remain in the fixed outcome denominator. Resource summaries report their available-
135 observation counts and missingness. Provider child, HTTP wrapper, gateway, scorer, and operator
136 clocks can overlap and are not added as disjoint costs. Shared native preparation is counted once.
137 Baseline checks, prior failed attempts, review, queue waits, cleanup, and artifact work retain their
138 own accounting scopes. Token usage and cost ticks are telemetry; unavailable settled charges and
139 unmeasured effort remain unknown.

140 The developmental pilot is a separate four-family, 24-cell A/B study with three nested repetitions. Its
141 retained labels include ten useful completions, five per arm, from 24 POSTs and 12 scores. Ten child
142 durations exceeded 595 seconds by less than 1 ms, and pilot cell 20 exceeded 600 seconds in both
143 wrapper and gateway clocks. These deviations and labels remain preserved; the final resource policy
144 is not applied retrospectively to relabel the pilot.

145 Earlier boundary evidence is also separate from final repair outcomes. The mutation/cold comparison
146 retains one false reuse among all 26 cases, rather than reporting a selected zero-of-25 subset. All four
147 later correction invocations, including failed and partial attempts, remain visible; corrected full-cold
148 execution supplies no positive reuse credit. Constructed gate checks and public-baseline passes do
149 not count as useful final repairs.

150 5 Results

151 All 32 planned cells are terminal. Useful completion is 9/32 (0.28125): arm A 5/16 and arm B 4/16.
152 Mean family B-minus-A useful fraction is -0.0625 . The frozen 20,000-resample whole-family
153 percentile interval is $[-0.1875, 0]$ and is not degenerate. The interval includes 0 and does not support
154 a superiority, noninferiority, equality, or safety claim. Table 1 and Figure 1 report the eight family
155 blocks. Tornado is the only family whose two-repetition means differ (A both useful; B one useful).
156 Tomlkit and more-itertools are tied; the other five families contribute zeros on both arms.

Table 1: Useful completions among two nested repetitions per family and arm. Families are the analysis blocks. Generated from the complete 32-cell analysis; repetitions are nested, not independent repositories.

Family	A useful / 2	B useful / 2	B-A mean
09-tomlkit	1	1	0
10-installer	0	0	0
11-tornado	2	1	-0.5
12-more-itertools	2	2	0
13-charset_normalizer	0	0	0
14-iniconfig	0	0	0
15-wheel	0	0	0
16-jinja	0	0	0
All eight (16 cells per arm)	5 / 16	4 / 16	-0.0625

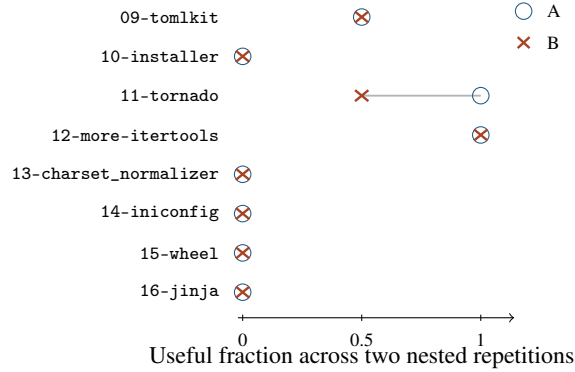


Figure 1: Eight family blocks: descriptive A/B useful fractions. Open circles are arm A; crosses are arm B. Coordinates match Table 1 and the generated eight-family analysis. This published figure is a Type 1 vector drawing of those coordinates. It is not the historical ASCII rendering retained as a prior-task generated artifact.

Outcomes on the fixed denominator are 9 `full_cold_pass`, 7 `hidden_acceptance_failed`, 14 `proposal_child_deadline`, and 2 `known_proposal_failure`. Sixteen cells reached scoring. Fifteen completed nonempty cold validation: nine useful passes and six executed hidden-check failures. The remaining scored cell (installer, arm A) preserved its signed `hidden_acceptance_failed` label after public checks passed 28/28 while hidden collection produced zero assertions and an import-name failure. That cell is incomplete validation, not an observed hidden assertion failure and not a proven external infrastructure fault. The other sixteen cells are the fourteen child-deadline overruns and two known proposal failures. No cell leaves the denominator.

Fourteen measured child-deadline overruns remain resource-unqualified, including submillisecond overshoots of the 595-second child bound. Nominal limits do not establish measured compliance. Cell 7 retains the original lease interruption; cells 8–32 use the separately admitted continuation runtime. Reconstructed zero-change candidate metadata on fifteen failed proposals is labeled derived sidecar data, not a returned model edit. One cell has no signed candidate and stays absent.

Cost and missing settlement. Provider-child, HTTP-wrapper, and gateway-proposal clocks are known for all 32 cells (known sums 13439.3, 13471.2, and 13488.6 seconds). Gateway-score clocks are known for 16/32 cells (known sum 83.394 seconds); the other 16 are missing, not zero. All 32 rows keep unknown external charge, and settlement is unavailable. Observed API usage on 16/32 cells includes 7125 completion tokens, 229853 prompt tokens, 532847 total tokens, and 7.30409×10^9 cost ticks; ticks are telemetry, not invoices. Grant-to-terminal, client-poll, and operator-phase clocks retain separate nested endpoints and are not added as one elapsed total. Isolated ablation cost components are null. Authoring effort, human time, and missing operator phases remain unmeasured.

Withdrawn endpoints and historical controls. Table 3 records the endpoints that this experiment does not measure. Empty ablation files are not zeros. Table 2 retains six historical boundary rows from constructed controls and scope closures. Identifiers of the form NS-NNN in that table are internal qualification IDs in the supplement evidence index; they are not families, not independent final trials, and not useful repairs. Public-baseline passes and constructed controls remain outside the 9/32 useful count.

The four-family developmental pilot remains a separate 24-cell study with retained labels of 10/24 useful completions, five per arm, from 24 POSTs and 12 scores. Its ten tiny child overruns and cell-20 wrapper/gateway overruns are not relabeled under the final policy. Historical false reuse remains 1/26; later correction invocations retain 48 attempted cases and 47 completed rows, with two 34-test public baselines and no reuse credit.

6 Related work

Proof-carrying code places explicit evidence between a producer and a consumer’s safety policy [4]. This motivates separating an artifact from the evidence needed to accept it. Our finite-check repair experiment and signed provenance records are not proof-carrying code in that formal sense. Abstract interpretation and translation validation similarly motivate explicit approximation and conversion relations [1, 5], not automatic admission of a backend or source-language proof.

Mokhov et al. separate scheduling from the decision whether a task needs rebuilding [2]. Our reuse obligation additionally names the statement, environment, lifecycle, and trust conditions for transferring a test observation. The cold comparison supplies no measured incremental-reuse advantage over build systems.

SWE-agent studies an interface for repository navigation, editing, and program execution [6]; SWT-Bench studies test-based validation of real-world fixes [3]. They motivate treating both information access and executable checks as parts of a repair method. Differences in tasks, interaction, models, and acceptance criteria preclude a head-to-head performance claim or an assertion that we reran their benchmarks. The contribution here is a bounded composition of evidence and authority mechanisms, not a first-of-kind claim or a new foundational algorithm.

A companion autoformalization study asks whether compiler-guided multi-view representations improve checked proof coverage; a companion law-to-action study asks whether source-grounded constraints prevent non-admitted tool effects. Those are different questions, oracles, and endpoints. Shared kernel, translation, and storage surfaces are one inspection each, not three independent implementations. Unavailable native proving is one environment fact, not three negative experiments.

7 Limitations

Eight convenience-selected families and two nested repetitions cannot support broad repository-population inference. The intervention changes the complete context supplied to the model; component ablations and a prompt-compression causal claim remain unavailable. The withdrawn C/D, warm-cache, reuse, publication, and sixteen-family objectives are not completed outcomes.

Finite tests cover their executed assertions. Passing them does not establish complete behavioral correctness, source-level semantic fidelity, or adequate coverage of every relevant behavior. Independent human semantic annotations, expert agreement, and human time were not measured. AI candidate inspection does not supply those measurements. Neither their absence nor an outside reviewer is a prerequisite for reporting the available automated evidence.

The conditional design arguments retain checker adequacy, complete dependency projections, protected policy authority, and exclusive checked publication as premises. They establish no universal Python theorem, native proof performance, global consensus, external exactly-once effects, or complete adversarial scorer integrity. A local constructed witness cannot fill a missing production or final-comparison measurement.

The interrupted runtime, actual timing deviations, prior failures, and unknown remote charges limit net-efficiency claims. Lease occupancy is not compute usage. Overlapping clocks and a pilot cost projection that aliases the same executions cannot be charged twice. Separate source epochs and actual cap observations must remain visible even when a candidate passes its tests.

Numerical reproduction from anonymous, manifest-bound scalar derivatives is narrower than original-signature verification or confidential-oracle rescoring. Changed derivative bytes are not authenticated by the original signature. Reference source derivatives must be distinguished from executed source, with upstream attribution and licences retained. Stable provenance hashes can permit correlation, so a lexical anonymity audit is bounded and does not guarantee de-identification.

8 Conclusion

On the frozen 32-cell A/B local-cold comparison, useful completion was 9/32 (A 5/16, B 4/16). The mean family B-minus-A useful fraction was -0.0625 , with descriptive conditional 95% interval $[-0.1875, 0]$. This is not a demonstrated improvement, equivalence, or safety claim. Fourteen child-

239 deadline overruns and seven hidden-acceptance failures, including one incomplete hidden collection,
240 remain in the denominator with two known proposal failures. Sixteen scoring invocations produced
241 fifteen completed cold validations. C/D routing, warm caches, reuse, publication efficacy, and sixteen-
242 family inference were withdrawn before outcomes and are not completed measurements. Human
243 semantic fidelity was unmeasured. Settled provider charges are unavailable. The contribution is the
244 tested evidence-boundary implementation and this bounded eight-family descriptive comparison, not
245 universal semantic soundness or a complete adversarial scorer-integrity result.

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A Full conditional argument record

The following audited statements are retained verbatim from the earlier formal argument record. Their Equation 2, Algorithm 1, and statement identifiers name that record; they are not additional final experiments or numbered algorithms in the main text. References to a sealed profile or the absence of a campaign describe the scope of that argument audit. Later constructed corrections and the cold comparison do not retroactively discharge its universal premises. The intended algorithm's bounded retry branch is architectural specification; the final empirical protocol permits no proposal retry.

Conditional acceptance and reuse arguments (audited)

The statements below restate Equation 2, the accepted-lineage sketch, the dependency-projection reuse obligation, the three acceptance questions, and Algorithm 1 with explicit scope, premises, and conclusions. Every sketch is **unmachine-checked**: the sealed validation profile has no usable `z3`, `lean`, `coqc`, or `cvc5`, and no independently checked proof artifact discharges these arguments. Implementation evidence supports only the named tested premises. External side-effect atomicity and universal dependency closure are not inferred. Unknown specifications stay visible.

Statement FA-EQ2 (acceptance invariant)

Scope. One publication of an accepted lineage root under a declared artifact policy. The statement is a design invariant for that publication operation. It is not a theorem that every current write path enforces the invariant, and it is not a claim about Git, remote stores, or physical effects outside the tested expected-parent compare-and-swap.

Premises. Let s be the current accepted root, s' a candidate successor, a the requesting actor, and q the required obligation set. Write $\text{Publish}(s')$ for the unique accepted-lineage publication of s' .

Conclusion.

$$\begin{aligned} \text{Publish}(s') \Rightarrow & \text{CurrentParent}(s) \wedge \text{Authorized}(a, s) \\ & \wedge \text{Scoped}(a) \wedge \text{CompleteManifest}(q, s') \\ & \wedge \text{EvidenceAdmitted}(q, s') \wedge \text{CAS}(s, s'). \end{aligned} \quad (2')$$

Argument (unmachine-checked). If publication is defined to succeed only after those six gates, then any published s' satisfies them. The converse is not claimed: a candidate that meets the gates may still lose compare-and-swap. Failed, stale, incomplete, or simulated attempts are not Publish successors.

What the invariant does not establish. Properties omitted by the checked policy; truth of unsupported or lossy translations; honesty of signed observations; atomicity of external side effects; global consensus; or universal enforcement by unexamined write paths.

Statement FA-I31 (accepted-lineage induction)

Scope. Roots reachable *solely* through accepted publication transitions of the operation in FA-EQ2, under a fixed statement profile Θ and policy predicate P_Θ over artifact manifests.

Premises.

1. **Initial-root policy.** The initial accepted root satisfies P_Θ .
2. **Exact input.** Required subject bytes and dependency references are verified before admission.
3. **Complete manifest.** The requirement manifest is complete under an admitted completeness rule; membership of some units is not completeness.
4. **Checker.** Each accepted evidence item satisfies its designated checker relation and evidence class.

- 309 5. **Translation bound.** Lossy or unsupported translations cannot discharge source-level
 310 obligations.
- 311 6. **Current parent.** Publication names the expected current parent.
- 312 7. **Exclusive publication.** Compare-and-swap linearizes concurrent same-parent writers; losers
 313 are not successors.
- 314 8. **Protected authority.** Policy and checker identities cannot change under ordinary worker
 315 authority.

316 **Conclusion.** Every root reachable solely through accepted publication transitions satisfies P_Θ .

317 **Argument (unmachine-checked).** Induction on the number of successful publications. The base
 318 case is premise 1. For the inductive step, the publisher checks P_Θ on the exact candidate manifest
 319 and current parent before compare-and-swap linearization. Failed, stale, incomplete, or simulated
 320 attempts are excluded from the successor relation, so they are not inductive cases. The induction
 321 preserves only properties actually expressed by P_Θ .

322 **Nonvacuity.** The successor relation is nonempty on the tested sealer: a complete current manifest
 323 can publish, while an empty or incomplete mandatory manifest cannot. Deny-all publication would
 324 make the invariant hold vacuously and is not a useful-progress witness.

325 **Statement FA-I32 (dependency-projected test reuse)**

326 **Scope.** Transfer of a previous setup/call/teardown observation of test t between states s and s' under
 327 a declared runtime/effect profile θ . The statement is a proof *obligation*, not a theorem of arbitrary
 328 pytest execution.

329 **Premises.** Let $\text{Obs}_\theta(t, s)$ be the relevant observation. Suppose there exist a complete projection
 330 π_D and a deterministic function f_θ such that

$$\text{Obs}_\theta(t, s) = f_\theta(\pi_D(s))$$

331 on every state admitted by the reuse policy.

332 **Conclusion.** If $\pi_D(s) = \pi_D(s')$, then $\text{Obs}_\theta(t, s) = \text{Obs}_\theta(t, s')$. An admitted previous observation
 333 may transfer only inside that profile and those assumptions.

334 **Argument (unmachine-checked).** Immediate from functionality of f_θ . The difficult premises
 335 are completeness and determinism of π_D . Equal digests of incomplete manifests do not satisfy π_D
 336 completeness. Opaque, uncontrolled, or unknown fixture values, incomplete runtime traces, and
 337 teardown failures do not discharge the obligation; they require execution or fail closed. A cold
 338 full-run oracle and adversarial mutations can check an implementation; they do not prove universal
 339 dependency closure. No such independent cold-oracle campaign is recorded here.

340 **Statement FA-I33 (three acceptance questions)**

341 **Scope.** Any candidate action that might be described as “accepted.”

342 **Premises and conclusions (separate).**

Question	What a positive answer establishes, and what it does not
Behavioral claim	The candidate program or model satisfies its stated obligation under the declared checker, environment, and evidence class. This is not permission to act and not publication of a lineage root.
343 Action permission	This actor may perform this action now under the bound lease, policy, scope, and roots. This is not behavioral correctness and not publication.
Publication	This candidate and its evidence become the current accepted root. This is not, by itself, proof that omitted obligations hold or that external effects occurred exactly once.

344 Neither a proof of behavior nor a capability grant answers all three. Formalization fixes the statement;
 345 authorization binds actor, tool, arguments, resources, effects, scope, time, and roots; execution
 346 observations report what happened; publication chooses the accepted lineage.

347 **Statement FA-ALG1 (composed supervision loop)**

348 **Scope.** Algorithm 1 remains the intended composition: a control-flow specification, not a newly
 349 executed full-loop result. It is labeled **executed composition** only on the steps supported by the
 350 integrated guard-before-write witness. Passing isolated stop-path tests does not establish the complete
 351 loop.

352 **Intended algorithm.** Input: authorized query q , current source forest F , policy P , budget B .

- 353 1. Reconcile event, task, and artifact state; refresh F and derive G, O .
- 354 2. If cancelled or budget exhausted: return explicit non-success.
- 355 3. If every current obligation is admitted: return goal-completion verification.
- 356 4. Generate bounded task proposals; preview and separately admit ready work.
- 357 5. If no task is admissible: return wait-or-review, not goal success.
- 358 6. Claim the task; bind lease/fence, preimages, required context, and validation.
- 359 7. Choose a deterministic, authorized residual-model, review, or unavailable route.
- 360 8. If review or unavailable: record a blocked attempt; return to the event service.
- 361 9. Produce and apply a candidate only through checked isolated execution.
- 362 10. Rescan; derive O_{new} from authorized rules, not the worker's pass claim.
- 363 11. Verify or reuse evidence; retain failed, unknown, and pending obligations.
- 364 12. If the failure is refinable: record a new residual; return a bounded retry.
- 365 13. If evidence is incomplete or unsafe: return defer-or-reject.
- 366 14. Persist the manifest; recheck current bindings; compare-and-swap the expected parent.
- 367 15. If publication conflicts: return reconcile, not completion.
- 368 16. Record the accepted task transition and updated F, G, O ; reevaluate at line 1.
- 369 17. Propose procedure or policy learning separately; never self-promote.

370 **Executed composition (integrated witness only).** The integrated witness supports a frozen guard-
 371 before-write cycle: authorized pre-state and obligation; counterexample; admitted deterministic route;
 372 isolated patch application; independent tests; narrow cache reuse of an unchanged eligible receipt;
 373 complete-manifest publication; and rejection of invalid, incomplete, and simulated non-completion.
 374 Those steps correspond to intended lines 4, 7 (deterministic), 9, 11 (narrow reuse), 13, 14, and 16,
 375 plus the negative reading of line 17 (no worker self-promotion). They do *not* execute the live event-
 376 service loop, residual production-model dispatch, budget-exhaustion composition, wait-or-review,
 377 procedure/policy learning, native proof, federation, or a matched A-D experiment.

378 **Trusted limitations retained with the algorithm.** Unknown or inconsistent specifications remain
 379 visible. Incomplete projections cannot silently authorize reuse. External exactly-once behavior
 380 outside the tested write-ahead-log / compare-and-swap / lifecycle boundary is not claimed. Optional
 381 native proving and world/procedure extensions are out of scope unless a later task supplies an eligible
 382 profile and consumed-result evidence.

383 **B Boundary controls and withdrawn endpoints**

384 Table 2 copies the six historical boundary rows. Identifiers NS-007 through NS-015, NS-027, and
 385 related NS-NNN tokens are internal qualification IDs defined by the supplement evidence index.
 386 They name constructed controls, public-baseline checks, or scope closures. They are not repository
 387 families, not independent final trials, and not useful repairs in the 32-cell comparison. Table 3 states
 388 withdrawn endpoints without converting absence into a zero effect.

Table 2: Historical boundary rows retained from constructed controls and scope closures. NS-NNN identifiers are internal qualification IDs, not families or independent final trials. Layout copy of the generated table with two allowbreak points only; scientific cells are otherwise unchanged.

Boundary	Safe rejection	Valid progress	Evidence
Provider gate	Missing/mismatched authority and untrusted close rejected in fixtures.	Deterministic repair and injected residual fixture; no production routing claim.	NS-008
Loss-aware translation	Modal erasure and unvalidated solver lift refused.	Exact/conservative translation only; no kernel source proof.	NS-009
Native theorem/circuit	Untested: no admitted native proof profile.	Unavailable: no genuine proof or measured prove/verify run.	NS-014 scope closure
Fixture/call/teardown reuse	Prior false reuse 1/26 retained; corrected route requires full cold.	Full public cold baseline only; no admitted positive reuse or speedup.	NS-011; corrected NS-027
Publication after restart	Empty/stale/failed publication refused in local controls.	WAL/CAS recovery and valid continuation; explicit hermetic shims.	NS-012/013
World/procedure consumption	Untested/out of scope.	Untested: no retained paired pre/post consumption.	NS-015 scope closure

Table 3: Explicit out-of-scope endpoints for the frozen A/B local-cold comparison. Empty ablation files are not zeros and do not estimate isolated component effects.

Endpoint	Scope
C/D routing/governance comparative benefits	Withdrawn or unmeasured; no final comparison or effect estimate.
proof/test reuse or local_warm effects	Withdrawn or unmeasured; no final comparison or effect estimate.
certificate/publication comparative benefits	Withdrawn or unmeasured; no final comparison or effect estimate.
adversarial scorer-integrity qualification	Withdrawn or unmeasured; no final comparison or effect estimate.
independent human semantic annotation	Withdrawn or unmeasured; no final comparison or effect estimate.
Non-A/B planned cells (128)	Withdrawn; no measured ablation rows.
A/B warm planned cells (32)	Withdrawn; no measured ablation rows.